

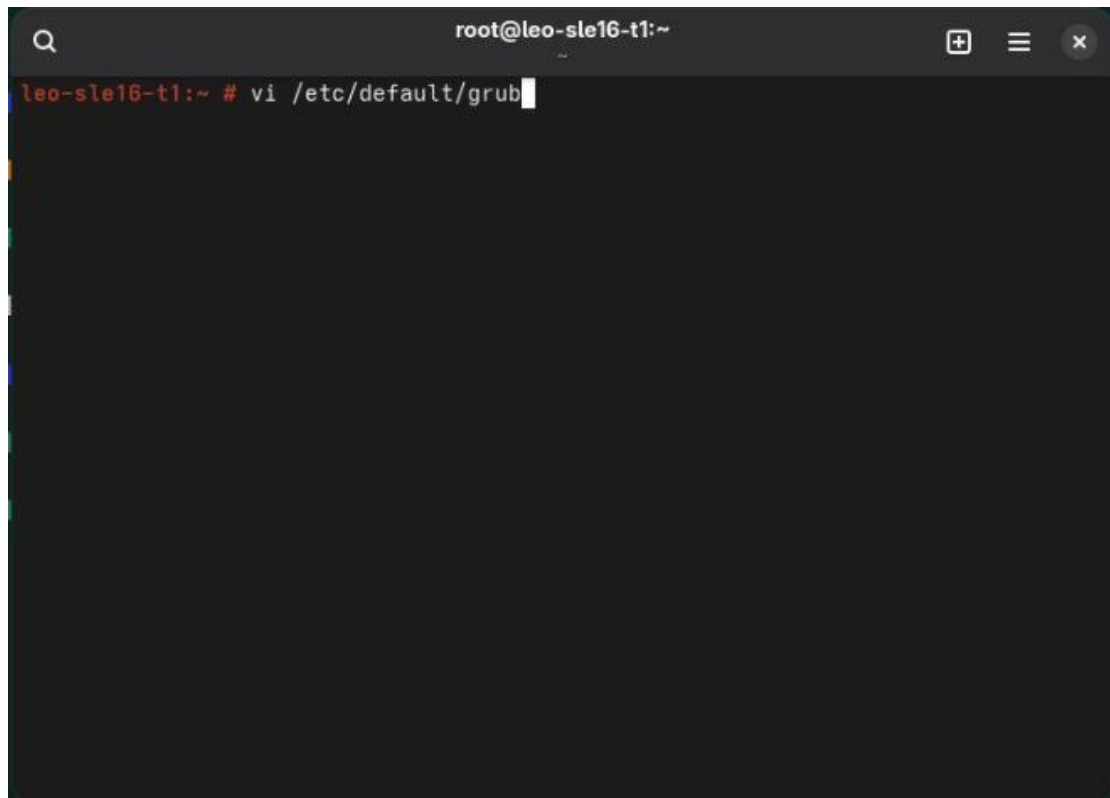
SLE16.0 SR-IOV setting

SLE16.0 SR-IOV 設定

1. 開啟虛擬化功能

編輯 GRUB 設定檔

#vim /etc/default/grub



A terminal window titled "root@leo-sle16-t1:~" with a search icon, a plus icon, a menu icon, and a close icon in the top right corner. The terminal shows the command "leo-sle16-t1:~ # vi /etc/default/grub" with a cursor at the end of the line.

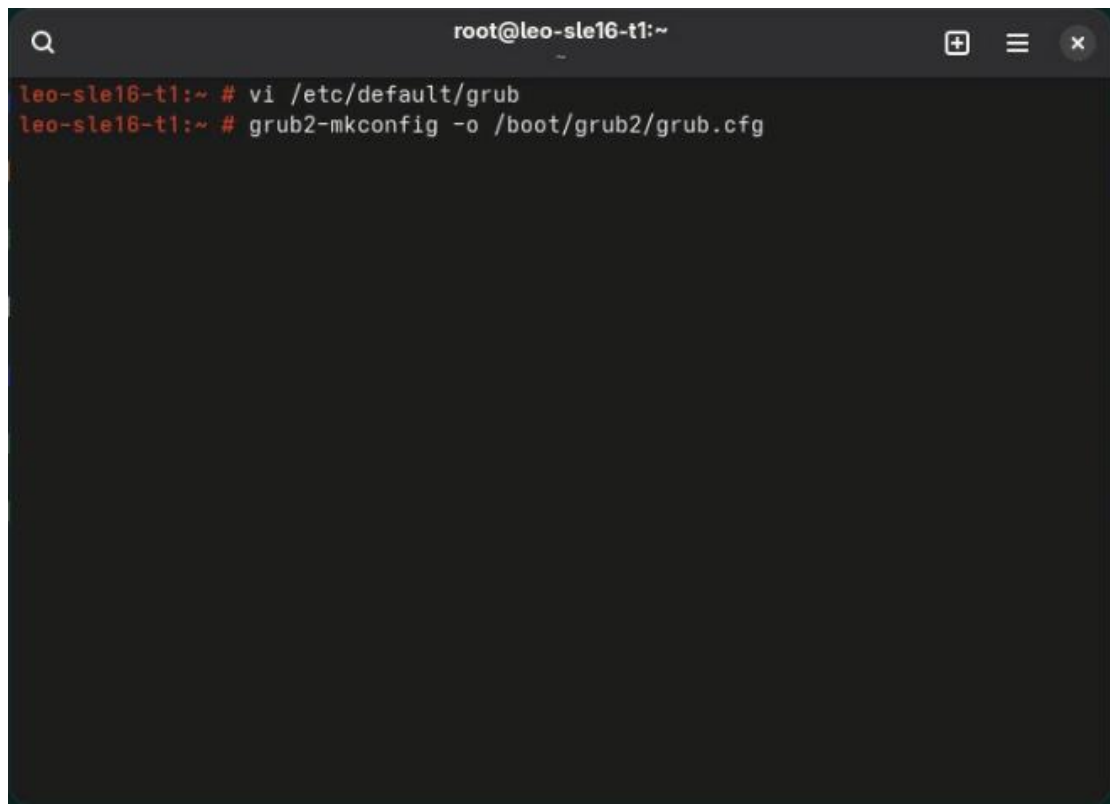
修改設定

在 GRUB_CMDLINE_LINUX_DEFAULT 後加入 'intel_iommu=on iommu=pt'

```
root@leo-sle16-t1:~  
# If you change this file, run 'grub2-mkconfig -o /boot/grub2/grub.cfg' afterwards to update  
# /boot/grub2/grub.cfg.  
  
# Uncomment to set your own custom distributor. If you leave it unset or empty, the default  
# policy is to determine the value from /etc/os-release  
GRUB_DISTRIBUTOR=  
GRUB_DEFAULT=saved  
GRUB_HIDDEN_TIMEOUT=0  
GRUB_HIDDEN_TIMEOUT_QUIET=true  
GRUB_TIMEOUT=8  
GRUB_CMDLINE_LINUX_DEFAULT="mitigations=auto quiet security=selinux selinux=1 crashkernel=72M,low crashkernel=404M,high intel_iommu=on iommu=pt"  
GRUB_CMDLINE_LINUX=""  
  
# Uncomment to automatically save last booted menu entry in GRUB2 environment  
  
# variable `saved_entry`  
# GRUB_SAVEDefault="true"  
# Uncomment to enable BadRAM filtering, modify to suit your needs  
  
# This works with Linux (no patch required) and with any kernel that obtains  
-- INSERT --  
11,145 Top
```

更新設定

```
#grub2-mkconfig -o /boot/grub2/grub.cfg
```

A terminal window titled 'root@leo-sle16-t1:~' with search, maximize, and close icons. It shows two commands being executed: 'vi /etc/default/grub' and 'grub2-mkconfig -o /boot/grub2/grub.cfg'.

```
root@leo-sle16-t1:~  
leo-sle16-t1:~ # vi /etc/default/grub  
leo-sle16-t1:~ # grub2-mkconfig -o /boot/grub2/grub.cfg
```

重開機套用設定

#reboot

```
root@leo-sle16-t1:~  
leo-sle16-t1:~ # vi /etc/default/grub  
leo-sle16-t1:~ # grub2-mkconfig -o /boot/grub2/grub.cfg  
Generating grub configuration file ...  
Found theme: /boot/grub2/themes/SLE/theme.txt  
Found linux image: /boot/vmlinuz-6.12.0-160000.5-default  
Found initrd image: /boot/initrd-6.12.0-160000.5-default  
Adding boot menu entry for UEFI Firmware Settings ...  
done  
leo-sle16-t1:~ # reboot
```

2.確認網路狀態

#ip link show

```
root@leo-sle16-t1:~  
setenforce: SELinux is disabled  
leo-sle16-t1:~ # ip link show  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
2: eno2np0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN mode DEFAULT group default qlen 1000  
    link/ether 14:23:f3:f1:49:12 brd ff:ff:ff:ff:ff:ff  
    altname enp81s0f0np0  
    altname enx1423f3f14912  
3: eno1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP mode DEFAULT group default qlen 1000  
    link/ether 30:e2:71:00:02:9e brd ff:ff:ff:ff:ff:ff  
    altname enp46s0  
    altname enx30e27100029e  
4: eno3np1: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN mode DEFAULT group default qlen 1000  
    link/ether 14:23:f3:f1:49:13 brd ff:ff:ff:ff:ff:ff  
    altname enp81s0f1np1  
    altname enx1423f3f14913  
5: ens3f0np0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP mode DEFAULT group default qlen 1000  
    link/ether 6c:fe:54:64:04:f8 brd ff:ff:ff:ff:ff:ff  
    altname enp173s0f0np0  
    altname enx6cfe546404f8  
6: ens3f1np1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP mode DEFAULT group default qlen 1000  
    link/ether 6c:fe:54:64:04:f9 brd ff:ff:ff:ff:ff:ff  
    altname enp173s0f1np1  
    altname enx6cfe546404f9  
7: ens3f2np2: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN mode DEFAULT group default qlen 1000  
    link/ether 6c:fe:54:64:04:fa brd ff:ff:ff:ff:ff:ff  
    altname enp173s0f2np2  
    altname enx6cfe546404fa  
8: ens3f3np3: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN mode DEFAULT group default qlen 1000  
    link/ether 6c:fe:54:64:04:fb brd ff:ff:ff:ff:ff:ff  
    altname enp173s0f3np3  
    altname enx6cfe546404fb
```

Create VF

```
root@leo-sle16-t1:~  
leo-sle16-t1:~ # echo 1 > /sys/class/net/ens3f0np0/device/sriov_numvfs
```

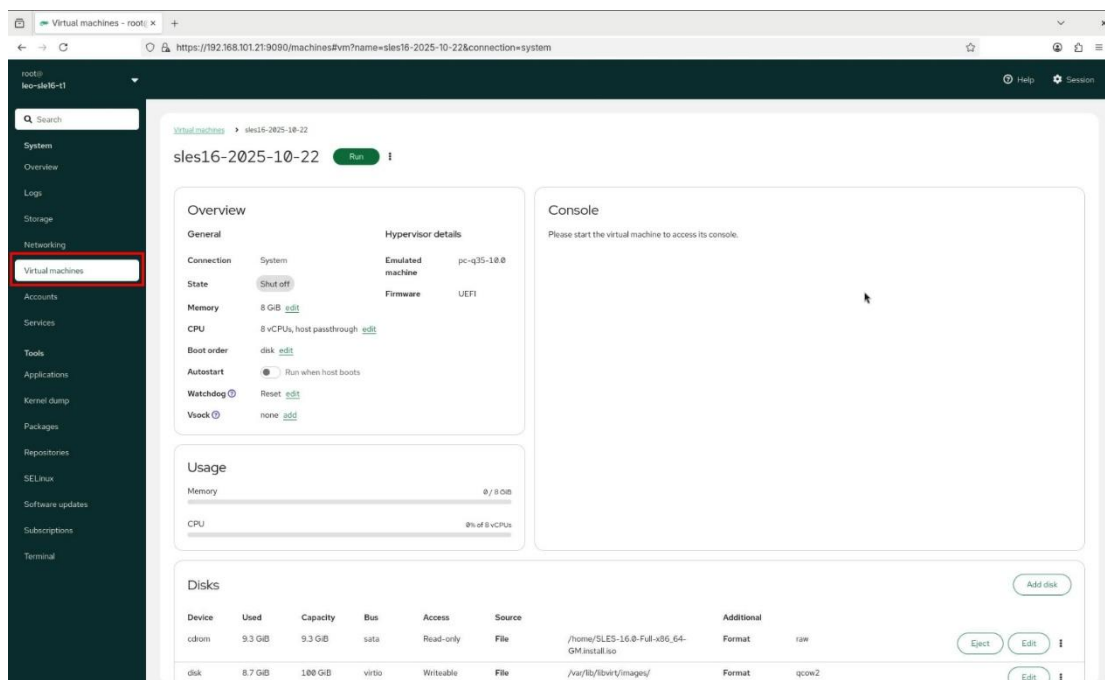
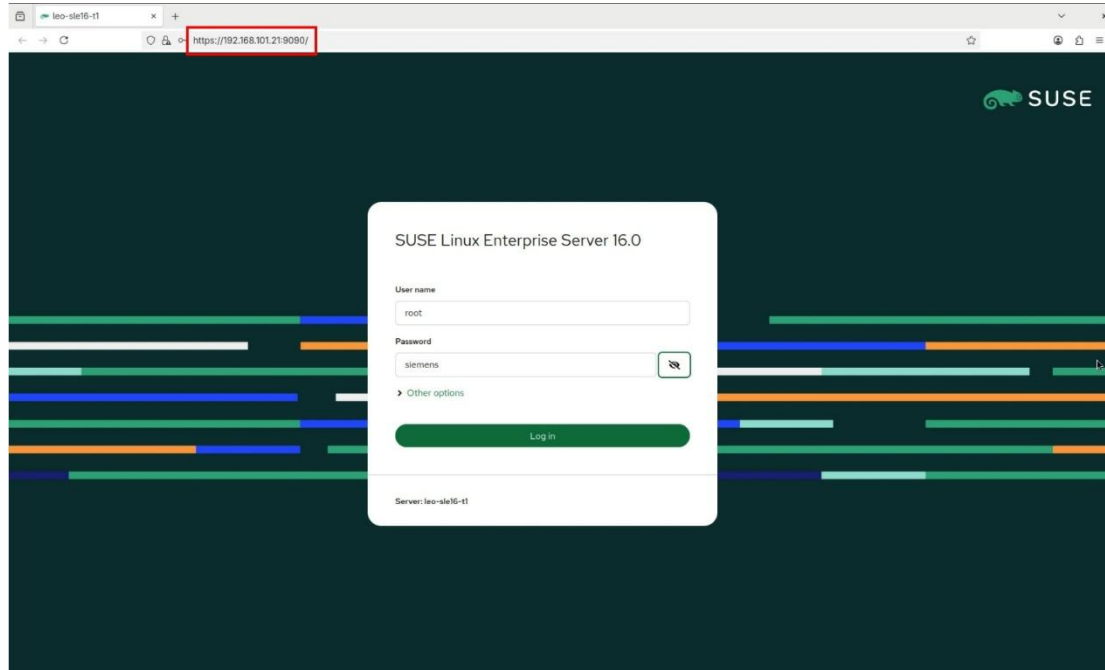
確認 VF 設定

#ip link show

```
root@leo-sle16-t1:~  
leo-sle16-t1:~ # echo 1 > /sys/class/net/ens3f0np0/device/sriov_numvfs  
leo-sle16-t1:~ # ip link show  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
2: eno2np0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN mode DEFAULT group default qlen 1000  
    link/ether 14:23:f3:f1:49:12 brd ff:ff:ff:ff:ff:ff  
    altname enp81s0f0np0  
    altname enx1423f3f14912  
3: eno1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP mode DEFAULT group default qlen 1000  
    link/ether 30:e2:71:00:02:9e brd ff:ff:ff:ff:ff:ff  
    altname enp46s0  
    altname enx30e27100029e  
4: eno3np1: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN mode DEFAULT group default qlen 1000  
    link/ether 14:23:f3:f1:49:13 brd ff:ff:ff:ff:ff:ff  
    altname enp81s0f1np1  
    altname enx1423f3f14913  
5: ens3f0np0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP mode DEFAULT group default qlen 1000  
    link/ether 6c:fe:54:64:04:f8 brd ff:ff:ff:ff:ff:ff  
    vf 0 link/ether 0e:c8:39:6c:e1:f5 brd ff:ff:ff:ff:ff:ff, spoof checking on, link-state auto, trust off  
    altname enp173s0f0np0  
    altname enx6cfe546404f8  
6: ens3f1np1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP mode DEFAULT group default qlen 1000  
    link/ether 6c:fe:54:64:04:f9 brd ff:ff:ff:ff:ff:ff  
    altname enp173s0f1np1  
    altname enx6cfe546404f9  
7: ens3f2np2: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN mode DEFAULT group default qlen 1000  
    link/ether 6c:fe:54:64:04:fa brd ff:ff:ff:ff:ff:ff  
    altname enp173s0f2np2  
    altname enx6cfe546404fa  
8: ens3f3np3: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN mode DEFAULT group default qlen 1000  
    link/ether 6c:fe:54:64:04:fb brd ff:ff:ff:ff:ff:ff  
    altname enp173s0f3np3  
    altname enx6cfe546404fb  
9: enp173s2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP mode DEFAULT group default qlen 1000  
    link/ether 0e:c8:39:6c:e1:f5 brd ff:ff:ff:ff:ff:ff  
leo-sle16-t1:~ #
```

3. 開啟 VM host ip + :9090

範例: 192.168.101.21:9090



Virtual machines - root | X

https://192.168.101.21:9090/machines?vm?name=sles16-2025-10-22&connection=system

HelpSession

root@leo-sles16-11

Search

System

Overview

Logs

Storage

Networking

Virtual machines

Accounts

Services

Tools

Applications

Kernel dump

Packages

Repositories

SELinux

Software updates

Subscriptions

Terminal

Memory

8 / 8 GiB

CPU

8% of 8 vCPUs

Disks

Add disk

Device	Used	Capacity	Bus	Access	Source	Additional	
cdrom	9.3 GiB	9.3 GiB	sata	Read-only	File	/home/SLES-16.0-Full+86_64-GM.install.iso	Eject Edit
disk	8.7 GiB	100 GiB	virtio	Writable	File	/my/lib/virt/images/sles16-2025-10-22.qcow2	Format Edit

Network interfaces

Add network interface

Type	Model type	MAC address	Source	State	
network	virtio	52:54:00:a7:ce:99	Network default	up	Unplug Edit

Host devices

Add host device

No host devices assigned to this VM

Snapshots

Create snapshot

No snapshots defined for this VM

Previously taken snapshots allow you to revert to an earlier state if something goes wrong

4. Add host device

選擇 PCI 勾選有'Virtual Function'選項加入虛擬機

Add host device

☐	(Undefined)	Intel Corporation	Slot	0000:fe:06.0
☐	(Undefined)	Intel Corporation	Slot	0000:fe:05.4
☒	Ethernet Virtual Function 700 Series	Intel Corporation	Slot	0000:ad:02.0
☐	(Undefined)	Intel Corporation	Slot	0000:7f:00.1
☐	(Undefined)	Intel Corporation	Slot	0000:fe:05.1
☐	(Undefined)	Intel Corporation	Slot	0000:fe:0f.0
☐	(Undefined)	Intel Corporation	Slot	0000:ff:00.0
☐	(Undefined)	Intel Corporation	Slot	0000:14:00.4
☐	(Undefined)	Intel Corporation	Slot	0000:6e:06.0
☐	(Undefined)	Intel Corporation	Slot	0000:ff:11.1
☐	(Undefined)	Intel Corporation	Slot	0000:fe:06.1
☐	(Undefined)	Intel Corporation	Slot	0000:fe:0e.0
☐	(Undefined)	Intel Corporation	Slot	0000:7e:00.2
☐	Ice Lake RAS	Intel	Slot	0000:5d:00.2

Add

Cancel

5. 進入系統後設定 Network

Cancel New Profile Add

Details Identity IPv4 IPv6 Security

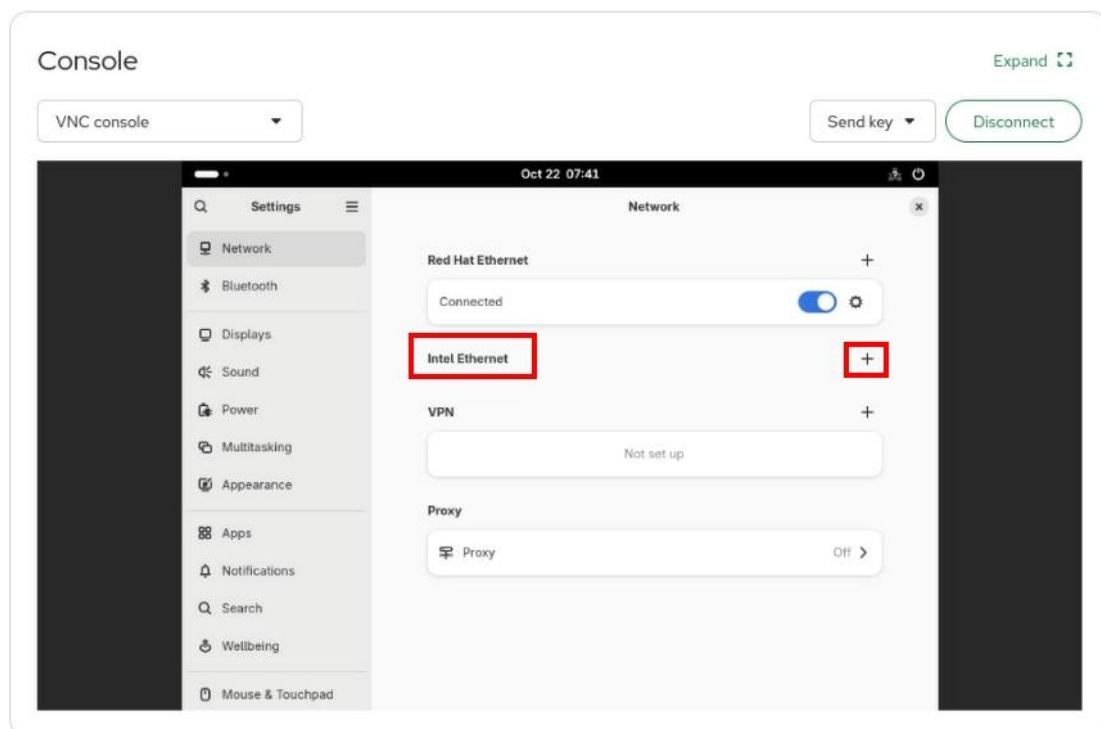
Name Profile 2

MAC Address 0E:C8:39:6C:E1:F5 (enp7s0)

Cloned Address

MTU automatic

Firewall Zone Default



Cancel

New Profile

Add

DetailsIdentityIPv4IPv6Security

IPv4 Method

☐ Automatic (DHCP)

☐ Link-Local Only

☒ Manual

☐ Disable

☐ Shared to other computers

Addresses

Address	Netmask	Gateway
192.168.1.3	24	

DNS

Automatic

Separate IP addresses with commas

Console

Expand

VNC console

Send key

Disconnect

Settings

Network

Bluetooth

Displays

Sound

Power

Multitasking

Appearance

Apps

Notifications

Search

Wellbeing

Mouse & Touchpad

Oct 22 07:43

Network

Red Hat Ethernet

Connected

Intel Ethernet

Connected - 10000 Mb/s

VPN

Not set up

Proxy

Proxy

Off